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CPSC 393

Prof. Ali

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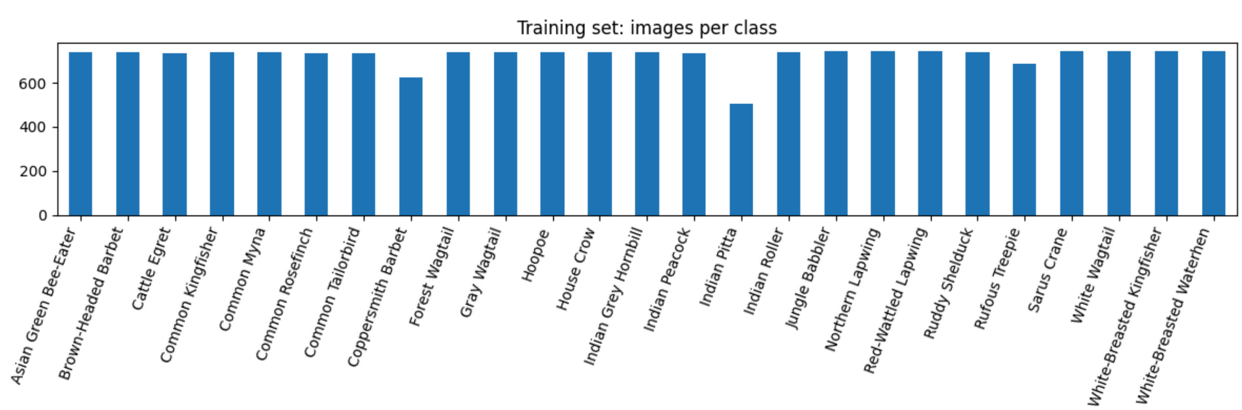
Technical Report - Homework Assignment 3

For this assignment, I developed two deep learning models to classify images of 25 Indian bird species, using a dataset composed of thousands of photographs collected across diverse environments. The goal of the project was to build and compare a custom Convolutional Neural Network (CNN) with a Transfer Learning model in order to evaluate how well each approach performs on a fine-grained visual recognition task. Because the dataset exhibits substantial variation in lighting, pose, background, and species similarity, the problem naturally lends itself to deep feature extraction methods. The custom CNN was trained from scratch to learn these visual patterns directly, while the Transfer Learning model leveraged a pretrained ResNet50 backbone to extract higher-level representations. By examining both architectures and analyzing their outputs, the project aimed to understand the strengths, limitations, and practical considerations of each model when applied to a complex real-world image classification challenge.

Analysis

For this assignment, I worked with a dataset containing 25 species of Indian birds, with over 200,000 images. All images were resized to 128×128 pixels and normalized to the range [0,1] to ensure consistent input formatting for both models.

Before building the models, I explored the dataset visually and numerically to understand class distributions and the overall difficulty of the task. A training-set histogram confirmed that most classes contained between 650–750 images, although a few, such as Indian Pitta, had closer to 500. While the imbalance was not extreme, it still has implications for learning, particularly because some of the underrepresented species are also visually subtle. The sample images generated during exploration highlighted the complexity of the problem: many photographs featured cluttered natural backgrounds, differences in lighting, and variations in the distance of the bird from the camera. Some birds were sharply captured with clear detail, while others were small, partially hidden, or surrounded by vegetation. Additionally, several species share similar colors or silhouettes, which makes this a fine-grained classification task. Overall, the EDA made it clear that this dataset would require models capable of extracting detailed features rather than relying on general shapes or simple textures.



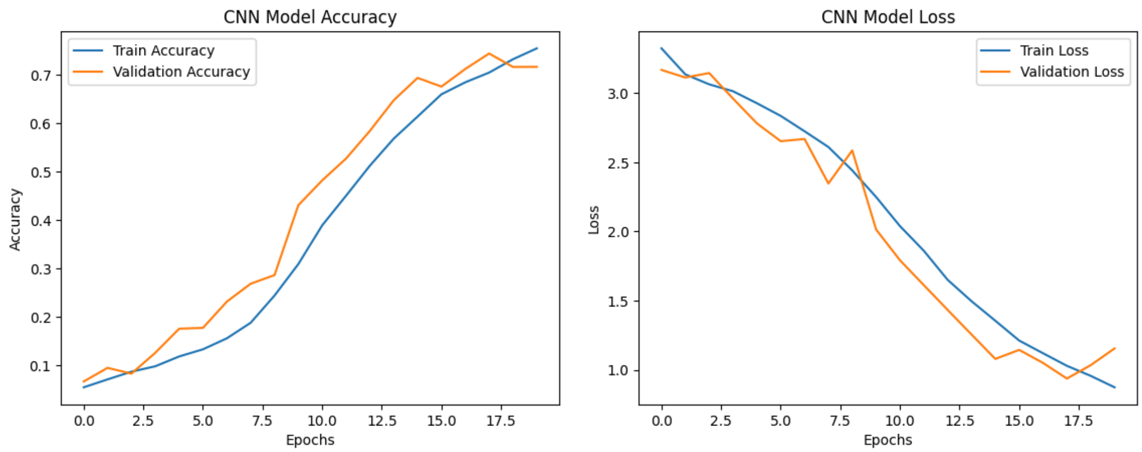
Methods

To address this classification challenge, I built two separate neural networks. The first was a deep Convolutional Neural Network (CNN). This model consisted of five convolutional layers paired with Batch Normalization and three MaxPooling operations to reduce spatial dimensions. As the network progressed in depth, the number of filters increased from 32 to 256, allowing the model to learn progressively more abstract features. After the convolutional blocks, the feature maps were flattened and passed into a dense layer with 256 neurons, followed by a Dropout layer with a 50% rate to reduce overfitting. The final output layer used softmax activation to classify the image into one of 25 bird species. The model was trained using the Adam optimizer and sparse categorical cross-entropy loss, which is appropriate for multi-class problems with integer labels.

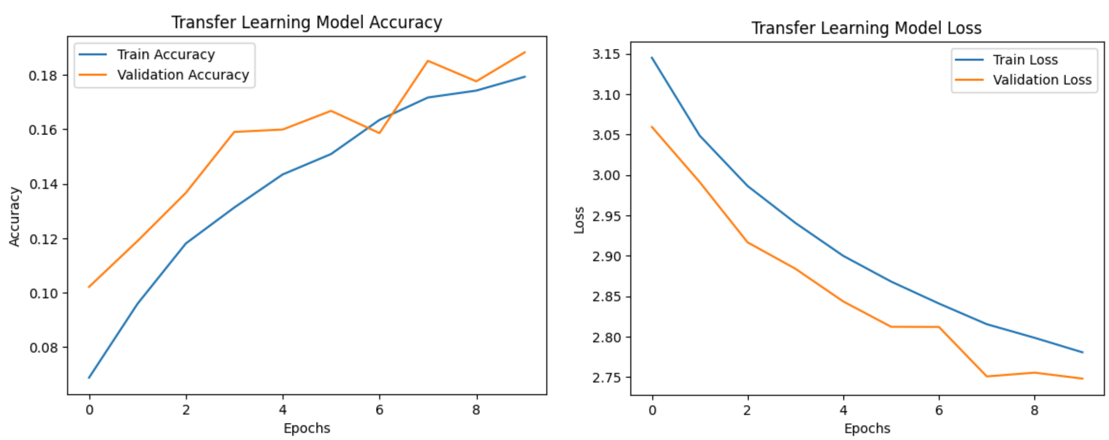
For the second model, I implemented a Transfer Learning approach using the ResNet50 architecture pretrained on ImageNet. I removed the fully connected layers, used the convolutional base as a fixed feature extractor, and added a custom classification head consisting of a Global Average Pooling layer, a dense layer of 256 units with ReLU activation, and a final softmax layer. All pretrained layers were frozen during training, meaning only the new head was updated. The reasoning behind this approach was to compare the performance of a well-designed custom CNN against a widely used pretrained model. However, because ResNet50 expects an input size closer to 224×224, and because I did not fine-tune the deeper layers, the transfer learning model might underperform on this fine-grained dataset.

Results

The CNN trained from scratch showed strong and consistent learning behavior. The accuracy curve demonstrated steady improvement over the 20 epochs, rising from roughly 6–8% in the first epoch to over 75% training accuracy by the end. The validation accuracy closely followed this trend, reaching approximately 72% and maintaining stable performance throughout the later epochs. The loss curve displayed a similarly healthy pattern, with both training and validation loss decreasing from around 3.2 at the beginning to approximately 0.88 (training) and 1.15 (validation) at the end. The proximity of the training and validation curves indicates that the regularization techniques were effective and prevented significant overfitting. Using the validation dataset as a test set, the final evaluation produced a test accuracy of 71.62% and a test loss of 1.1559, confirming solid generalization.



By contrast, the Transfer Learning model struggled noticeably. Although validation accuracy increased slightly from about 10% to 18% over 10 epochs, the improvement was shallow, and the model never showed signs of learning class-specific distinctions. The loss fell only gradually, from roughly 3.1 to 2.74, and did not exhibit the sharp downward trajectory seen in the CNN. The near-parallel behavior of training and validation accuracy also suggests that the model simply did not have the capacity to adapt meaningfully, and freezing the entire backbone prevented it from learning the visual cues necessary for distinguishing these bird species. The final accuracy of 18.83% was only slightly above random guessing, confirming that the transfer learning setup was not suitable for this task under the given constraints.



A brief review of misclassified cases from the CNN suggests that most errors occurred in situations where species shared highly similar color schemes, where birds appeared very small in the frame, or where backgrounds were particularly dense with foliage. Images showing unusual poses or partial obstructions also frequently led to misclassification. These observations reinforce the importance of clear subject visibility and pose consistency in fine-grained image classification tasks.

Reflection

This assignment provided valuable insight into both custom feature learning and transfer learning approaches. One of the most important lessons was that Transfer Learning is not automatically superior; its success depends heavily on appropriate preprocessing, input resolution, and whether fine-tuning is performed. In this case, using a completely frozen ResNet50 backbone limited the model’s ability to specialize to the subtle differences between bird species. Meanwhile, the CNN I built from scratch performed well, ultimately achieving over 70% accuracy without the benefit of pretraining.

Some challenges I faced included low initial accuracy during early experiments, training instability before adding Batch Normalization, and the computational demands of training multiple deep models. Additionally, analyzing the dataset revealed how factors such as class imbalance, background clutter, and variation in bird size influence learning outcomes. If I were to revisit this assignment, I would incorporate stronger data augmentation, explore fine-tuning the upper layers of ResNet50, experiment with higher input resolutions for the transfer model, and conduct more systematic error analysis to identify patterns directly tied to specific species.

Overall, this project deepened my understanding of convolutional architectures, the importance of regularization, and the nuanced considerations involved when choosing between training from scratch and leveraging pretrained models.